

Entity-Relationship Model Description

The database schema represents an International Airport Management System designed to handle flight operations, passenger bookings, and security logistics.

Entities and Attributes:

- Airports:** Stores airport details including `airport\_id` (PK), name, location (country, state, city), and audit timestamps.
- Airlines companies:** Stores airline details including `airline\_id` (PK), code, name, country, and audit timestamps.
- Actual flights information:** The central entity for flight logistics. It includes `flight\_id` (PK), departure/arrival gates, scheduled/actual times, and Foreign Keys linking to `Airlines companies` and `Airports` (for both departure and arrival locations).
- Passengers:** Stores personal data including `passenger\_id` (PK), name, gender, DOB, citizenship, residence, and passport number.
- Bookings:** Links passengers to flights. Attributes include `booking\_id` (PK), status, platform, price, and Foreign Keys for `flight\_id` and `passenger\_id`.
- Booking_Changes:** An audit table tracking modifications to bookings, linked via `booking\_id`.
- Boarding passes:** Stores issuance details including `boarding\_pass\_id` (PK), seat, boarding time, and a Foreign Key to `Bookings`.
- Registered baggage:** Tracks luggage details including `baggage\_id` (PK), weight, and a Foreign Key to `Bookings`.
- Checked baggage:** Stores security screening results for registered baggage. It includes `baggage\_checking\_id` (PK) and Foreign Keys to both `Bookings` and `Passengers`.
- Security check:** Tracks passenger screening. Includes `security\_check\_id` (PK), results, and a Foreign Key to `Passengers`.

Relationships and Cardinality:

One-to-Many:

One Airline operates many Flights

One Airport serves as the departure or arrival point for many Flights

One Passenger can make many Bookings

One Passenger can undergo many Security Checks

One Flight can have many Bookings

One Booking can have multiple Baggage items and one Boarding Pass

Many-to-One:

Many Bookings are associated with one Flight

Many Baggage items belong to one Booking

Conclusion

In this laboratory work, an Entity-Relationship Diagram (ERD) was successfully developed to model the operations of an International Airport. The diagram effectively captures the complex interactions between passengers, flight scheduling, airline management, and security protocols.

The database design normalizes the data to reduce redundancy while ensuring data integrity through the use of Primary and Foreign Key constraints. The schema supports critical airport operations, including:

1. ****Flight Management:**** Tracking scheduled versus actual flight times and gate assignments.
2. ****Passenger Processing:**** Managing bookings, issuing boarding passes, and recording security check results.
3. ****Baggage Handling:**** Distinguishing between simple baggage registration and security-checked baggage results.

The resulting model provides a solid foundation for a functional database system that can handle the day-to-day data requirements of a modern airport, ensuring efficient retrieval and management of information.

